CHAPTER 6 Integration

This chapter serves two purposes. It contains a careful development of the Riemann integral, which is the integral studied in standard calculus courses. It also contains an introduction to a generalization of the Riemann integral called the Riemann-Stieltjes integral. The generalization is easy and natural. Moreover, the Riemann-Stieltjes integral is an important tool in probability and statistics, and other areas of mathematics.

§32 The Riemann Integral

The theory of the Riemann integral is no more difficult than several other topics dealt with in this book. The one drawback is that it involves some technical notation and terminology.

32.1 Definition.
Let f be a bounded function on a closed interval $[a, b]$.[1] For $S \subseteq [a, b]$, we adopt the notation

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[1]Here and elsewhere in this chapter, we assume $a < b$.

K.A. Ross, *Elementary Analysis: The Theory of Calculus*,
Undergraduate Texts in Mathematics, DOI 10.1007/978-1-4614-6271-2_6,

$$M(f,S) = \sup\{f(x) : x \in S\} \quad \text{and} \quad m(f,S) = \inf\{f(x) : x \in S\}.$$

A *partition of* $[a,b]$ is any finite ordered subset P having the form

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}.$$

The *upper Darboux sum* $U(f,P)$ of f with respect to P is the sum

$$U(f,P) = \sum_{k=1}^{n} M(f,[t_{k-1},t_k]) \cdot (t_k - t_{k-1})$$

and the *lower Darboux sum* $L(f,P)$ is

$$L(f,P) = \sum_{k=1}^{n} m(f,[t_{k-1},t_k]) \cdot (t_k - t_{k-1}).$$

Note

$$U(f,P) \leq \sum_{k=1}^{n} M(f,[a,b]) \cdot (t_k - t_{k-1}) = M(f,[a,b]) \cdot (b-a);$$

likewise $L(f,P) \geq m(f,[a,b]) \cdot (b-a)$, so

$$m(f,[a,b]) \cdot (b-a) \leq L(f,P) \leq U(f,P) \leq M(f,[a,b]) \cdot (b-a). \quad (1)$$

The *upper Darboux integral* $U(f)$ of f over $[a,b]$ is defined by

$$U(f) = \inf\{U(f,P) : P \text{ is a partition of } [a,b]\}$$

and the *lower Darboux integral* is

$$L(f) = \sup\{L(f,P) : P \text{ is a partition of } [a,b]\}.$$

In view of (1), $U(f)$ and $L(f)$ are real numbers.

We will prove in Theorem 32.4 that $L(f) \leq U(f)$. This is not obvious from (1). [Why?] We say f is *integrable* on $[a,b]$ provided $L(f) = U(f)$. In this case, we write $\int_a^b f$ or $\int_a^b f(x)\,dx$ for this common value:

$$\int_a^b f = \int_a^b f(x)\,dx = L(f) = U(f). \quad (2)$$

Specialists call this integral the *Darboux integral.* Riemann's definition of the integral is a little different [Definition 32.8], but we will show in Theorem 32.9 that the definitions are equivalent. For this

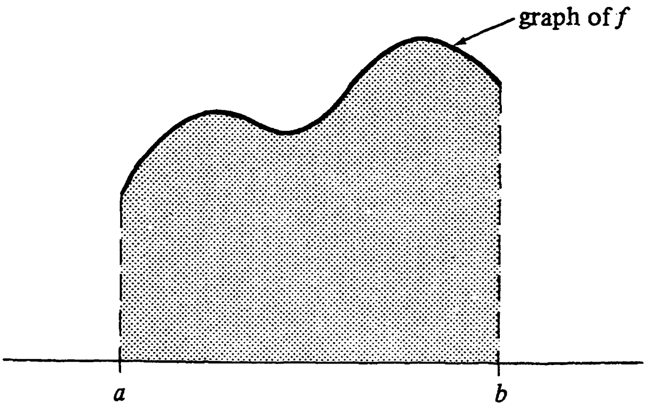

FIGURE 32.1

reason, we will follow customary usage and call the integral defined above the *Riemann integral*.

For nonnegative functions, $\int_a^b f$ is interpreted as the area of the region under the graph of f [see Fig. 32.1] for the following reason. Each lower Darboux sum represents the area of a union of rectangles inside the region, and each upper Darboux sum represents the area of a union of rectangles that contains the region. Moreover, $\int_a^b f$ is the unique number that is larger than or equal to all lower Darboux sums and smaller than or equal to all upper Darboux sums. Figure 19.2 on page 145 illustrates the situation for $[a,b] = [0,1]$ and

$$P = \left\{0 < \frac{1}{n} < \frac{2}{n} < \cdots < \frac{n-1}{n} < 1\right\}.$$

Example 1

The simplest function whose integral is not obvious is $f(x) = x^2$. Consider f on the interval $[0,b]$ where $b > 0$. For a partition

$$P = \{0 = t_0 < t_1 < \cdots < t_n = b\},$$

we have

$$U(f,P) = \sum_{k=1}^{n} \sup\{x^2 : x \in [t_{k-1}, t_k]\} \cdot (t_k - t_{k-1}) = \sum_{k=1}^{n} t_k^2 (t_k - t_{k-1}).$$

If we choose $t_k = \frac{kb}{n}$, then we can use Exercise 1.1 to calculate

$$U(f, P) = \sum_{k=1}^{n} \frac{k^2 b^2}{n^2}\left(\frac{b}{n}\right) = \frac{b^3}{n^3}\sum_{k=1}^{n} k^2 = \frac{b^3}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}.$$

For large n, this is close to $\frac{b^3}{3}$, so we conclude $U(f) \leq \frac{b^3}{3}$. For the same partition we find

$$L(f, P) = \sum_{k=1}^{n} \frac{(k-1)^2 b^2}{n^2}\left(\frac{b}{n}\right) = \frac{b^3}{n^3} \cdot \frac{(n-1)(n)(2n-1)}{6},$$

so $L(f) \geq \frac{b^3}{3}$. Therefore $f(x) = x^2$ is integrable on $[0, b]$ and

$$\int_0^b x^2\,dx = \frac{b^3}{3}.$$

Of course, any calculus student could have calculated this integral using a formula that is based on the Fundamental Theorem of Calculus; see Example 1 in §34. □

Example 2
Consider the interval $[a, b]$, where $a < b$, and let $f(x) = 1$ for rational x in $[a, b]$, and let $f(x) = 0$ for irrational x in $[a, b]$. For any partition

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\},$$

we have

$$U(f, P) = \sum_{k=1}^{n} M(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) = \sum_{k=1}^{n} 1 \cdot (t_k - t_{k-1}) = b - a$$

and

$$L(f, P) = \sum_{k=1}^{n} 0 \cdot (t_k - t_{k-1}) = 0.$$

It follows that $U(f) = b - a$ and $L(f) = 0$. The upper and lower Darboux integrals for f do not agree, so f is not integrable! □

We next develop some properties of the integral.

32.2 Lemma.
Let f be a bounded function on $[a, b]$. If P and Q are partitions of $[a, b]$ and $P \subseteq Q$, then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P). \tag{1}$$

Proof
The middle inequality is obvious. The proofs of the first and third inequalities are similar, so we will prove

$$L(f, P) \leq L(f, Q). \tag{2}$$

An induction argument [Exercise 32.4] shows we may assume
Q has only one more point, say u, than P. If

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\},$$

then

$$Q = \{a = t_0 < t_1 < \cdots < t_{k-1} < u < t_k < \cdots < t_n = b\}$$

for some $k \in \{1, 2, \ldots, n\}$. The lower Darboux sums for P and Q are the same except for the terms involving t_{k-1} or t_k. In fact, their difference is

$$\begin{aligned} L(f, Q) - L(f, P) = m(f, [t_{k-1}, u]) \cdot (u - t_{k-1}) + m(f, [u, t_k]) \cdot (t_k - u) \\ - m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}). \end{aligned} \tag{3}$$

To establish (2) it suffices to show this quantity is nonnegative. Using Exercise 4.7(a), we see

$$\begin{aligned} & m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) \\ & \quad = m(f, [t_{k-1}, t_k]) \cdot \{(t_k - u) + (u - t_{k-1})\} \\ & \quad \leq m(f, [u, t_k]) \cdot (t_k - u) + m(f, [t_{k-1}, u]) \cdot (u - t_{k-1}). \end{aligned}$$

■

32.3 Lemma.
If f is a bounded function on $[a, b]$, and if P and Q are partitions of $[a, b]$, then $L(f, P) \leq U(f, Q)$.

Proof
The set $P \cup Q$ is also a partition of $[a, b]$. Since $P \subseteq P \cup Q$ and $Q \subseteq P \cup Q$, we can apply Lemma 32.2 to obtain

$$L(f, P) \leq L(f, P \cup Q) \leq U(f, P \cup Q) \leq U(f, Q).$$

■

32.4 Theorem.
If f is a bounded function on $[a,b]$, then $L(f) \le U(f)$.

Proof
Fix a partition P of $[a,b]$. Lemma 32.3 shows $L(f,P)$ is a lower bound for the set

$$\{U(f,Q) : Q \text{ is a partition of } [a,b]\}.$$

Therefore $L(f,P)$ is less than or equal to the greatest lower bound [infimum!] of this set. That is

$$L(f,P) \le U(f). \tag{1}$$

Now (1) shows that $U(f)$ is an upper bound for the set

$$\{L(f,P) : P \text{ is a partition of } [a,b]\},$$

so $U(f) \ge L(f)$. ■

Note that Theorem 32.4 also follows from Lemma 32.3 and Exercise 4.8; see Exercise 32.5. The next theorem gives a "Cauchy criterion" for integrability.

32.5 Theorem.
A bounded function f on $[a,b]$ is integrable if and only if for each $\epsilon > 0$ there exists a partition P of $[a,b]$ such that

$$U(f,P) - L(f,P) < \epsilon. \tag{1}$$

Proof
Suppose first that f is integrable and consider $\epsilon > 0$. There exist partitions P_1 and P_2 of $[a,b]$ satisfying

$$L(f,P_1) > L(f) - \frac{\epsilon}{2} \quad \text{and} \quad U(f,P_2) < U(f) + \frac{\epsilon}{2}.$$

For $P = P_1 \cup P_2$, we apply Lemma 32.2 to obtain

$$\begin{aligned} U(f,P) - L(f,P) &\le U(f,P_2) - L(f,P_1) \\ &< U(f) + \frac{\epsilon}{2} - \left[L(f) - \frac{\epsilon}{2}\right] = U(f) - L(f) + \epsilon. \end{aligned}$$

Since f is integrable, $U(f) = L(f)$, so (1) holds.

Conversely, suppose for $\epsilon > 0$ the inequality (1) holds for some partition P. Then we have

$$\begin{aligned} U(f) \leq U(f,P) &= U(f,P) - L(f,P) + L(f,P) \\ &< \epsilon + L(f,P) \leq \epsilon + L(f). \end{aligned}$$

Since ϵ is arbitrary, we conclude $U(f) \leq L(f)$. Hence we have $U(f) = L(f)$ by Theorem 32.4, i.e., f is integrable. ■

The remainder of this section is devoted to establishing the equivalence of Riemann's and Darboux's definitions of integrability. Subsequent sections will depend only on items Definition 32.1 through Theorem 32.5. Therefore the reader who is content with the Darboux integral in Definition 32.1 can safely proceed directly to the next section.

32.6 Definition.
The *mesh* of a partition P is the maximum length of the subintervals comprising P. Thus if

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\},$$

then

$$\text{mesh}(P) = \max\{t_k - t_{k-1} : k = 1, 2, \ldots, n\}.$$

Here is another "Cauchy criterion" for integrability.

32.7 Theorem.
A bounded function f on $[a,b]$ is integrable if and only if for each $\epsilon > 0$ there exists a $\delta > 0$ such that

$$\text{mesh}(P) < \delta \quad \textit{implies} \quad U(f,P) - L(f,P) < \epsilon \tag{1}$$

for all partitions P of $[a,b]$.

Proof
The ϵ–δ condition in (1) implies integrability by Theorem 32.5.

Conversely, suppose f is integrable on $[a,b]$. Let $\epsilon > 0$ and select a partition

$$P_0 = \{a = u_0 < u_1 < \cdots < u_m = b\}$$

of $[a,b]$ such that

$$U(f,P_0) - L(f,P_0) < \frac{\epsilon}{2}. \tag{2}$$

Since f is bounded, there exists $B > 0$ such that $|f(x)| \le B$ for all $x \in [a,b]$. Let $\delta = \frac{\epsilon}{8mB}$; m is the number of intervals comprising P_0.

To verify (1), we consider any partition

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}$$

with $\text{mesh}(P) < \delta$. Let $Q = P \cup P_0$. If Q has one more element than P, then a glance at (3) in the proof of Lemma 32.2 leads us to

$$L(f,Q) - L(f,P) \le B \cdot \text{mesh}(P) - (-B) \cdot \text{mesh}(P) = 2B \cdot \text{mesh}(P).$$

Since Q has at most m elements that are not in P, an induction argument shows

$$L(f,Q) - L(f,P) \le 2mB \cdot \text{mesh}(P) < 2mB\delta = \frac{\epsilon}{4}.$$

By Lemma 32.2 we have $L(f,P_0) \le L(f,Q)$, so

$$L(f,P_0) - L(f,P) < \frac{\epsilon}{4}.$$

Similarly

$$U(f,P) - U(f,P_0) < \frac{\epsilon}{4},$$

so

$$U(f,P) - L(f,P) < U(f,P_0) - L(f,P_0) + \frac{\epsilon}{2}.$$

Now (2) implies $U(f,P) - L(f,P) < \epsilon$ and we have verified (1). ■

Now we give Riemann's definition of integrability.

32.8 Definition.
Let f be a bounded function on $[a,b]$, and let $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ be a partition of $[a,b]$. A *Riemann sum* of f associated with the partition P is a sum of the form

$$\sum_{k=1}^{n} f(x_k)(t_k - t_{k-1})$$

where $x_k \in [t_{k-1}, t_k]$ for $k = 1, 2, \ldots, n$. The choice of x_k's is quite arbitrary, so there are infinitely many Riemann sums associated with a single function and partition.

The function f is *Riemann integrable* on $[a, b]$ if there exists a number r with the following property. For each $\epsilon > 0$ there exists $\delta > 0$ such that

$$|S - r| < \epsilon \tag{1}$$

for every Riemann sum S of f associated with a partition P having $\text{mesh}(P) < \delta$. The number r is the *Riemann integral* of f on $[a, b]$ and will be provisionally written as $\mathcal{R}\int_a^b f$.

32.9 Theorem.
A bounded function f on $[a, b]$ is Riemann integrable if and only if it is [Darboux] integrable, in which case the values of the integrals agree.

Proof
Suppose first that f is [Darboux] integrable on $[a, b]$ in the sense of Definition 32.1. Let $\epsilon > 0$, and let $\delta > 0$ be chosen so that (1) of Theorem 32.7 holds. We show

$$\left| S - \int_a^b f \right| < \epsilon \tag{1}$$

for every Riemann sum

$$S = \sum_{k=1}^{n} f(x_k)(t_k - t_{k-1})$$

associated with a partition P having $\text{mesh}(P) < \delta$. Clearly we have $L(f, P) \le S \le U(f, P)$, so (1) follows from the inequalities

$$U(f, P) < L(f, P) + \epsilon \le L(f) + \epsilon = \int_a^b f + \epsilon$$

and

$$L(f, P) > U(f, P) - \epsilon \ge U(f) - \epsilon = \int_a^b f - \epsilon.$$

This proves (1); hence f is Riemann integrable and

$$\mathcal{R}\int_a^b f = \int_a^b f.$$

Now suppose f is Riemann integrable in the sense of Definition 32.8, and consider $\epsilon > 0$. Let $\delta > 0$ and r be as given in Definition 32.8. Select any partition

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}$$

with $\mathrm{mesh}(P) < \delta$, and for each $k = 1, 2, \ldots, n$, select x_k in $[t_{k-1}, t_k]$ so that

$$f(x_k) < m(f, [t_{k-1}, t_k]) + \epsilon.$$

The Riemann sum S for this choice of x_k's satisfies

$$S \le L(f, P) + \epsilon(b - a)$$

as well as

$$|S - r| < \epsilon.$$

It follows that

$$L(f) \ge L(f, P) \ge S - \epsilon(b - a) > r - \epsilon - \epsilon(b - a).$$

Since ϵ is arbitrary, we have $L(f) \ge r$. A similar argument shows $U(f) \le r$. Since $L(f) \le U(f)$, we see $L(f) = U(f) = r$. This shows f is [Darboux] integrable and

$$\int_a^b f = r = \mathcal{R}\int_a^b f.$$

■

32.10 Corollary.
Let f be a bounded Riemann integrable function on $[a, b]$. Suppose (S_n) is a sequence of Riemann sums, with corresponding partitions P_n, satisfying $\lim_n \mathrm{mesh}(P_n) = 0$. Then the sequence (S_n) converges to $\int_a^b f$.

Proof
Let $\epsilon > 0$. There is a $\delta > 0$ so that if S is a Riemann sum with corresponding partition P, and if $\text{mesh}(P) < \delta$, then

$$\left| S - \int_a^b f \right| < \epsilon.$$

Choose N so that $\text{mesh}(P_n) < \delta$ for $n > N$. Then

$$\left| S_n - \int_a^b f \right| < \epsilon \quad \text{for} \quad n > N.$$

Since $\epsilon > 0$ is arbitrary, this shows $\lim_n S_n = \int_a^b f$. ∎

32.11 Remark.
I recently had occasion to use the following simple observation. If one ignores the end intervals of the partitions, the "almost Riemann sums" so obtained still converge to the integral; see [59]. This arose because the intervals had the form $[a, b]$, but the partition points had the form $\frac{k}{n}$. Thus the partition points were nice and equally spaced, *except* for the end ones.

Exercises

32.1 Find the upper and lower Darboux integrals for $f(x) = x^3$ on the interval $[0, b]$. *Hint*: Exercise 1.3 and Example 1 in §1 will be useful.

32.2 Let $f(x) = x$ for rational x and $f(x) = 0$ for irrational x.

(a) Calculate the upper and lower Darboux integrals for f on the interval $[0, b]$.

(b) Is f integrable on $[0, b]$?

32.3 Repeat Exercise 32.2 for g where $g(x) = x^2$ for rational x and $g(x) = 0$ for irrational x.

32.4 Supply the induction argument needed in the proof of Lemma 32.2.

32.5 Use Exercise 4.8 to prove Theorem 32.4. Specify the sets S and T in this case.

32.6 Let f be a bounded function on $[a,b]$. Suppose there exist sequences (U_n) and (L_n) of upper and lower Darboux sums for f such that $\lim(U_n - L_n) = 0$. Show f is integrable and $\int_a^b f = \lim U_n = \lim L_n$.

32.7 Let f be integrable on $[a,b]$, and suppose g is a function on $[a,b]$ such that $g(x) = f(x)$ except for finitely many x in $[a,b]$. Show g is integrable and $\int_a^b f = \int_a^b g$. *Hint*: First reduce to the case where f is the function identically equal to 0.

32.8 Show that if f is integrable on $[a,b]$, then f is integrable on every interval $[c,d] \subseteq [a,b]$.

§33 Properties of the Riemann Integral

In this section we establish some basic properties of the Riemann integral and we show many familiar functions, including piecewise continuous and piecewise monotonic functions, are Riemann integrable.

A function is *monotonic* on an interval if it is either increasing or decreasing on the interval; see Definition 29.6.

33.1 Theorem.
Every monotonic function f on $[a,b]$ is integrable.

Proof
We assume f is increasing on $[a,b]$ and leave the decreasing case to Exercise 33.1. We also assume $f(a) < f(b)$, since otherwise f would be a constant function. Since $f(a) \le f(x) \le f(b)$ for all $x \in [a,b]$, f is clearly bounded on $[a,b]$. In order to apply Theorem 32.5, let $\epsilon > 0$ and select a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ with mesh less than $\frac{\epsilon}{f(b)-f(a)}$. Then

$$\begin{aligned} U(f,P) - L(f,P) &= \sum_{k=1}^{n} \{M(f,[t_{k-1},t_k]) - m(f,[t_{k-1},t_k])\} \cdot (t_k - t_{k-1}) \\ &= \sum_{k=1}^{n} [f(t_k) - f(t_{k-1})] \cdot (t_k - t_{k-1}). \end{aligned}$$